SUBJECTIVE WORKSHEET

Time and Motion

Science — Physics · Class 7 · Max Marks: 34 · Time: 1 hour

Section A — Very Short Answer (1 × 5 = 5)

1. State the SI unit of speed. [1]

2. Name the device used to measure the speed of a vehicle. [1]

3. Define the time period of a pendulum. [1]

4. Write the formula for speed. [1]

5. Name the type of motion shown by a swinging pendulum. [1]

Section B — Short Answer (2 × 5 = 10)

1. A train covers 150 km in 3 hours. Calculate its speed. [2]

2. Convert 90 km/h into metres per second. [2]

3. Give two differences between uniform and non-uniform motion. [2]

4. What is a simple pendulum? Name the factor that decides its time period. [2]

5. How can the speed of a body be found from its distance–time graph? [2]

Section C — Long Answer (5 × 3 = 15)

1. A body covers 60 m in the first 4 s, 40 m in the next 4 s and 80 m in the next 4 s. Is its motion uniform or non-uniform? Find the total distance, total time and average speed. [5]

2. Describe how a simple pendulum can be used to measure time, defining oscillation and time period. [5]

3. With an example of each, explain the distance–time graph of a body (i) at rest, (ii) in uniform motion and (iii) in non-uniform motion. [5]

Section D — Higher-Order Thinking (Give Reasons) (2 × 2 = 4)

1. Two cars start from the same point in the same direction; car A at 40 km/h and car B at 60 km/h. How far apart are they after 2 hours? [2]

2. If the thread of a pendulum is shortened, will each oscillation take more or less time? Give a reason. [2]

Answer Key

Very Short Answer

1. Metre per second (m/s).
2. Speedometer.
3. The time taken to complete one oscillation.
4. Speed = distance ÷ time.
5. Periodic (oscillatory) motion.

Short Answer

1. Speed = distance ÷ time = 150 ÷ 3 = 50 km/h.
2. 90 km/h = 90 × (1000/3600) = 90 × 5/18 = 25 m/s.
3. In uniform motion equal distances are covered in equal times (speed constant); in non-uniform motion unequal distances are covered in equal times (speed changes).
4. A simple pendulum is a small heavy bob hung by a light thread from a fixed point. Its time period depends on the length of the thread.
5. By taking the ratio of the distance covered to the time taken (the slope of the line): speed = distance ÷ time.

Long Answer

1. Unequal distances in equal times, so the motion is non-uniform. Total distance = 60 + 40 + 80 = 180 m; total time = 12 s; average speed = 180 ÷ 12 = 15 m/s.
2. A pendulum swings to and fro regularly. One complete to-and-fro movement is an oscillation, and the time for one oscillation is the time period. Since each oscillation of a given pendulum takes the same time, counting oscillations lets us measure time (as in a pendulum clock).
3. At rest — a horizontal line (distance does not change). Uniform motion — a straight slanting line (equal distances in equal times). Non-uniform motion — a curved line (unequal distances in equal times).

Higher-Order Thinking (Give Reasons)

1. A covers 80 km and B covers 120 km, so they are 120 - 80 = 40 km apart.
2. Less time — the time period decreases when the length of the pendulum is reduced.